Empirical IO – Problem Set 1: Top paid apps on Google Play (US)

Giovanni Cecchi, Zoe Di Lieto, David Herskovits, Lorenzo Morandi, Jakub Przewoski

October 2, 2026

1 Data and sample attrition

In **Step 1** we scraped all 500 apps. In **Step 2** we matched 357 apps via the `google-play-scraper`. 143 apps were dropped. Our scraper recorded the error for each dropped observation: `App not found (404)`, which corresponds to the app being removed from Google Play. We decided to eliminate these from the dataset that would be used in Problem Set 2 because no data was available (probaly due to outdated IDs), therefore they would just add noise to the estimates. **Finally,** we dropped 27 apps which had prices as 0. We decided to drop them because estimating a demand model with 0 price apps would make little sense, and they are not paid apps as they are free. So, we have 330 apps in total. We kept all the relevant features that the scraper function returned.

2 Quantity

To build the conservative measure of quantity, first we used the assumption that, regardless of the individual characteristics of the apps, each one would get one review for every 20 downloads. However, for 97% of apps, this measure would lead us to have fewer downloads than the lower bound for the number of downloads for each app ("MinInstalls"): if the constructed new quantity variable would be lower than the lower bound, then for the quantity value, we impute the lower bound itself. (We would impute the next lower bound - we will call it the upper bound - if the estimate would be higher than that but this is never the case.)

A better approach would be to use a midpoint between intervals instead of the lower bound if reviews are uninformative, assuming that e.g. 1000000+ downloads mean downloads between 1 and 5 million, 5 million being the next threshold (e.g. a geometric mean for simplicity, assuming log-uniform distribution). However, if we use reviews for prediction, then we can't use the geometric mean for the observations that do not reach the lower bound, because then we would estimate more downloads for an app that just didn't reach the threshold (the midpoint) than to an app that just did.

A slightly better approach would be assuming that there is one rating per every 20 downloads. This way, we would get an estimate lower than the lower bound for about 40% of the apps, and this time, for about 7% of the apps, the ceiling is binding (so based on our prediction, the range of installs should have been the next band). We adjust these for our imputation. But ratings are more informative than reviews, and the mean is now about 480 thousand higher than with the previous approach. With an even larger multiplier (30), we estimate another 290 thousand extra downloads, and now we do not reach the lower bound for about 25% of the apps, but we estimate more than the ceiling for 20% of the apps!

One way to use the midpoint while exploiting variation in ratings would be to average the midpoint (geometric mean of the lower and upper bounds of the intervals, taking as upper bound the next lower bound) and the previous estimate (30*ratings). This way, the lower bound needs to be used only about 6% of the time, and the upper bound 7%, so this is a better approach if we want to use the ceilings and floors as little as possible.

When scraping google play, one of the variables that result from the scraping is "realInstalls", whose name suggests that it represents a precise number of downloads for each app. Since this information is not public and we have no justification nor official documentation about it, we can't assume that it is completely reliable. At the same time, it seems to be a decent proxy, when compared to other ones that we construct.

Note that the closest imputation to realInstall is taking the average of the geometric mean and the ratings with the larger multiplier: the realInstall mean is about 1.246M installs, while our 4 estimation approaches lead us to estimates of around 715 thousand, 1.198M, 1.486M, and 1.447M installs on average, respectively.

See Figures 1 and 2 in the Appendix for plotting the scraped installs variable against the imputation using reviews and the imputation using ratings.

3 Genre groups and outside good

To build the outside option, we focused on the genre of the apps. By using the variable *genre_id* we aggregated non-game app genres that served to people similar services and constructed six different genres: Tools and Personalization; Productivity and Business; Photography and Video Players. Games still accounted for 60 per cent of the sample and a single "Games" group would treat very different titles as equally close substitutes. Therefore we split games into four groups: Puzzle & Strategy (puzzle, board, card, strategy, trivia), Action & Adventure (action, adventure, arcade, racing, sports), Role Playing, and Simulation. All apps that had a *genre_id* variable with no clear substitutes in different groups and genres counting less than 10 apps would be categorized as the outside good. The five largest groups are Puzzle and Strategy games (65), Action and Adventure games (63), Tools and Personalization (35), Simulation games (33), and Role Playing games (23). Moreover, we had 60 outside options.

4 Summary statistics

Table 1: Summary statistics, matched apps

	Mean	Median	SD	Min	Max	N
Price (USD)	5.66	4.49	6.93	0.49	84.99	330
Quantity (reviews)	715090.42	100000.00	2998958.97	1000.00	50000000.00	330
Quantity (ratings)	1197560.30	248850.00	5836109.69	1000.00	100000000.00	330
Quantity (ratings + [larger multiplier])	1486469.85	373275.00	6180141.10	1170.00	100000000.00	330
Quantity (ratings + installs midpoint)	1447291.34	298440.90	6075229.40	1703.03	100000000.00	330
Average score	4.22	4.32	0.53	1.97	4.97	330
Number of ratings	61158.00	11221.50	339230.24	39.00	5849116.00	330
Number of reviews	6354.09	1303.00	36707.88	8.00	635025.00	330
Installs (lower bound)	714700.00	100000.00	2999039.74	1000.00	50000000.00	330
In-app purchases (share)	0.35	0.00	0.48	0.00	1.00	330
Rank in top 500	245.05	241.50	146.16	1.00	500.00	330
Installs (scraped)	1345801.39	349645.00	5955860.96	2271.00	98108280.00	330

This table reports the summmary statistics of the apps for which information was available in Google Play. The table shows that the Price variable (USD) is on average low (5.66 USD), while there are still some outliers with a high cost that reach 84.99 USD.

The table also reports the four measures of quantity that we created. It is possible to see that the baseline proxy "quantity (downloads)" mainly reduces to the Google Play installs buckets. In fact the rows for the baseline quantity and the installs lower bound are practically identical: they have the same median (100,000), minimum and maximum, and their means differ by fewer than 500 downloads.

The alternative measures based on ratings give medians about three times larger (from 248,850 to 373,275), suggesting that the baseline proxy might be underestimating the number of downloads.

5 Price and quantity

The best-fitting line appears to be downward-sloping, albeit not steeply. Relatedly, apps with a relatively low number of downloads are not priced more highly compared to other more expensive products (crucially, abstracting from the apps' genre). It is evident that apps at the very extremes of the quantity range have very similar prices. Removing such regions from the range of downloads, the negative price-quantity demanded relationship is indeed a lot more evident. We observe the same patterns whether we use the scraped estimates for the number of installs, or our imputations based on reviews or ratings.

For the plots, please see Figures 3 and 4 in the Appendix.

Appendix: Figures

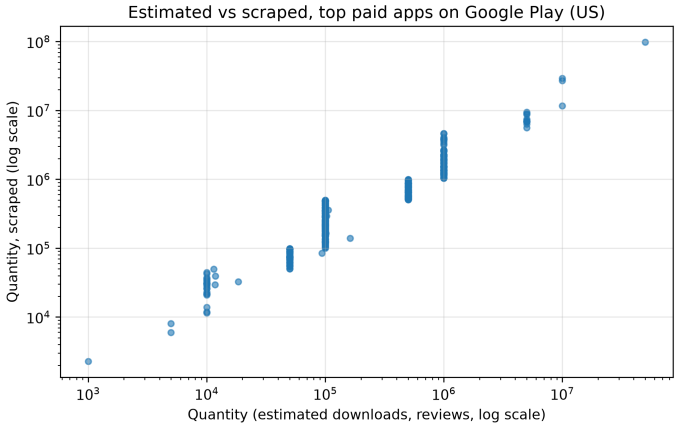


Figure 1: RealInstalls variable and imputed installs based on 20 installs per review

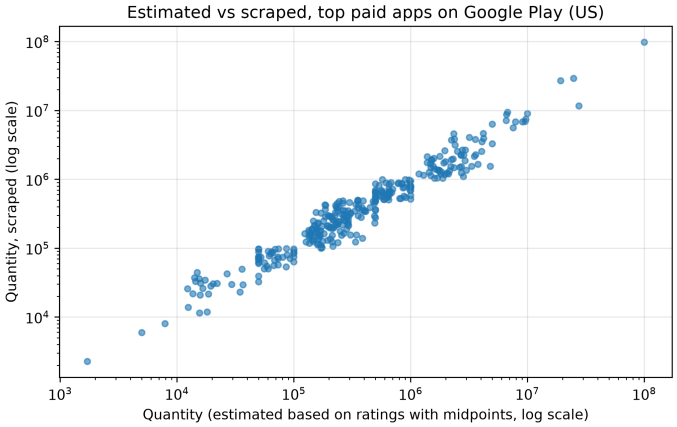


Figure 2: RealInstalls variable and imputed installs based on 20 installs per rating and midpoint

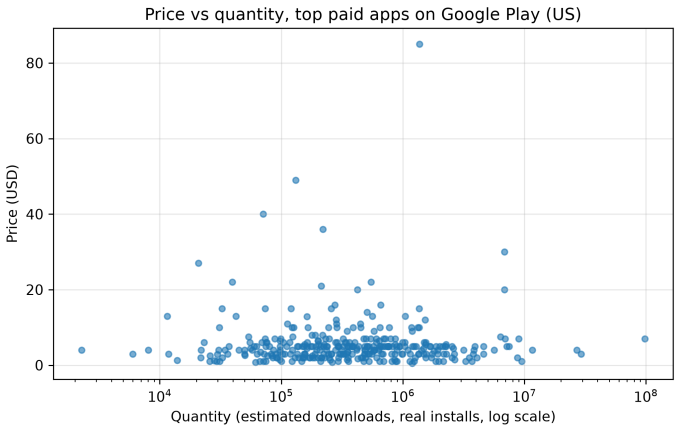


Figure 3: Price and RealInstalls variable

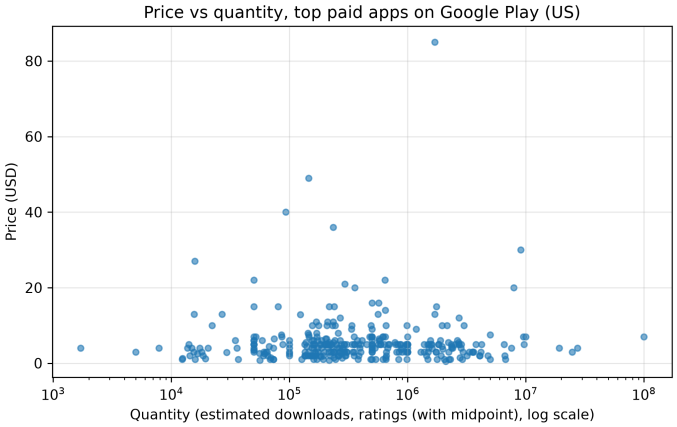


Figure 4: Price and imputed installs based on ratings of the interval and midpoint